Seven Results of the Vacuum Condensate: From Nuclear Matter to Quantum Mechanics

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Abstract

Within the framework of Null-Vector Gravity (NVG), we show that the vacuum condensate $\Phi = \mathcal{W}\,e^{i\theta}$, decomposed into amplitude $\mathcal{W}$ and Goldstone phase θ via the Madelung representation, provides a unified setting in which seven physical results can be derived from the single action (2) without introducing additional quantum postulates beyond the NVG Lagrangian: **(I)** the Heisenberg uncertainty principle as a Cauchy–Schwarz inequality for the fields $(\nabla \log \mathcal{W}, \nabla\theta)$; **(II)** the Born rule as the equilibrium distribution of localized excitations under osmotic drift and diffusion; **(III)** wave-function collapse as θ-phase thermalization with a characteristic timescale $\tau \sim \hbar/k_B T$ in the overdamped regime; **(IV)** the vanishing of the QCD vacuum angle $\bar{\theta}_{\mathrm{QCD}} = 0$ from the joint $\mathcal{W}$–θ potential minimum, structurally distinct from the Peccei–Quinn mechanism; **(V)** the thermodynamic arrow of time from the topological sector $Q = +1$; **(VI)** a conditional prediction that Bell–CHSH violation vanishes above the QCD deconfinement temperature $T_c = 157$ MeV, contingent on the NVG identification of photon polarization with the θ-field (testable at RHIC BES-II); and **(VII)** the Hawking temperature and Bekenstein–Hawking entropy as θ-mode thermalization on the horizon in classical curved spacetime. Numerical verification scripts reproducing every figure in this paper are available at `https://github.com/infosave2007/vmf`.

Contents

1 Introduction

The vacuum condensate field

$$\Phi(x) \;=\; \mathcal{W}(x)\, \mathrm{e}^{\mathrm{i}\theta(x)}\,, \tag{1}$$

where $\mathcal{W}(x) > 0$ is the condensate amplitude and $\theta(x)$ is the Goldstone phase, was introduced in Refs. [1, 2] as the single additional dynamical degree of freedom of Null-Vector Gravity (NVG). The full action reads

$$S = \int \mathrm{d}^4x\, \sqrt{-g}\left[\frac{R}{16\pi G} + \frac{1}{2}\,\partial_\mu\mathcal{W}\,\partial^\mu\mathcal{W} + \frac{\mathcal{W}^2}{2}\,\partial_\mu\theta\,\partial^\mu\theta - V(\mathcal{W},\theta)\right], \tag{2}$$

where the potential $V(\mathcal{W},\theta)$ possesses a global minimum at $(\mathcal{W},\theta) = (W_0,\, 0)$ with $W_0 = M_{\Omega,0} = 859$ MeV, and respects the approximate $U(1)$ symmetry $\theta \to \theta + \text{const}$, explicitly broken only by the QCD axial anomaly.

In preceding publications [1, 2, 3, 4, 5] we have demonstrated that NVG reproduces the nuclear equation of state anchored to lattice QCD sigma terms, neutron star masses and radii consistent with NICER data, a bounce cosmology avoiding the Big Bang singularity, and the neutrino mass spectrum from a single parameter f_a.

The purpose of the present paper is to show that the *same* object (1) yields seven results — here called "consequences" for brevity — that address foundational problems of physics within the NVG framework, without introducing new quantum postulates beyond the action (2). We emphasize that the action itself, including the potential $V(\mathcal{W},\theta)$ and the vacuum value $W_0 = M_{\Omega,0} = 859$ MeV, constitutes the defining assumption of NVG; what we show is that no *further* postulates (projection, Born rule, second law, Peccei–Quinn symmetry, etc.) are needed once (2) is accepted. The logical dependence is summarized in Table 1.

Table 1. Seven open problems of physics and their potential resolution via the vacuum condensate $\Phi = \mathcal{W}\,\mathrm{e}^{\mathrm{i}\theta}$. The column "Standard Physics" indicates the conventional status (postulate, unsolved, or separate mechanism); "NVG" states the result of this work.

#	Problem	Standard Physics	NVG Result
I	Heisenberg uncertainty	Postulate	Cauchy–Schwarz for $(\nabla \log \mathcal{W},\ \nabla\theta)$
II	Born rule $P = \lvert\psi\rvert^2$	Postulate	Osmotic drift-diffusion balance
III	Wave-function collapse	Projection postulate	θ-thermalization, $\tau = \hbar/k_B T$
IV	Strong CP: $\bar\theta = 0$	Axion / Peccei–Quinn	$\partial V/\partial\theta\big\rvert_{W_0,0} = 0$
V	Arrow of time	Second law (postulate)	Topological charge $Q = +1 > 0$
VI	Bell nonlocality	"Spooky action"	θ-coherence; $S \to 0$ for $T > T_c$
VII	Quantum gravity	Unresolved (strings, LQG, . . .)	θ-thermalization on horizon

2 Result I: Heisenberg Uncertainty Principle

Result 1 (Heisenberg uncertainty as Cauchy–Schwarz)**.** *Let $\rho(x) = \mathcal{W}(x)^2$ define a probability density and introduce the $L^2(\rho)$ inner product $\langle f, g\rangle_\rho \equiv \int f(x)\, g(x)\, \rho(x)\,\mathrm{d}x$. Then*

$$\boxed{\Delta x \cdot \Delta p \ \geq\ \frac{\hbar}{2}} \tag{3}$$

is a consequence of the Cauchy–Schwarz inequality applied to the amplitude gradient $f = \nabla \log \mathcal{W}$ and the shifted position coordinate $g = x - \langle x\rangle_\rho$.

Proof. In the Madelung representation [6], $\psi = \sqrt{\rho}\,\mathrm{e}^{\mathrm{i}S/\hbar}$ maps onto the NVG condensate $\Phi = \mathcal{W}\,\mathrm{e}^{\mathrm{i}\theta}$ via $\rho = \mathcal{W}^2$ and $S = \hbar\theta$. The position variance encodes the spread of the amplitude:

$$(\Delta x)^2 = \langle (x - \langle x\rangle_\rho)^2\rangle_\rho\,. \tag{4}$$

The momentum variance of the wave function $\psi = \mathcal{W}\,\mathrm{e}^{\mathrm{i}\theta}$ reads:

$$\begin{aligned}(\Delta p)^2 &= \langle \hat p^2\rangle - \langle \hat p\rangle^2 \\ &= \hbar^2\left[\langle (\nabla \log \mathcal{W})^2\rangle_\rho + \langle (\nabla\theta)^2\rangle_\rho - \langle \nabla\theta\rangle_\rho^2\right] \\ &\geq \hbar^2 \langle (\nabla \log \mathcal{W})^2\rangle_\rho\,,\end{aligned} \tag{5}$$

where we used $\langle \hat p\rangle = \hbar\langle\nabla\theta\rangle_\rho$ and the fact that the phase variance $\langle(\nabla\theta)^2\rangle_\rho - \langle\nabla\theta\rangle_\rho^2 \geq 0$.

Applying the Cauchy–Schwarz inequality to $f = \nabla \log \mathcal{W}$ and $g = x - \langle x\rangle_\rho$ in $L^2(\rho)$ yields:

$$\langle (\nabla \log \mathcal{W})^2\rangle_\rho \ \cdot\ \langle (x - \langle x\rangle_\rho)^2\rangle_\rho \ \geq\ \Big|\langle \nabla \log \mathcal{W} \cdot (x - \langle x\rangle_\rho)\rangle_\rho\Big|^2. \tag{6}$$

Using $\nabla \log \mathcal{W} = \nabla\mathcal{W}/\mathcal{W}$ and $\rho = \mathcal{W}^2$, the cross-term simplifies via integration by parts:

$$\langle \nabla \log \mathcal{W} \cdot (x - \langle x\rangle_\rho)\rangle_\rho = \int_{-\infty}^{\infty} \mathcal{W}\,\nabla\mathcal{W} \cdot (x - \langle x\rangle_\rho)\,\mathrm{d}x = \frac{1}{2}\int_{-\infty}^{\infty} \nabla(\mathcal{W}^2)\cdot (x - \langle x\rangle_\rho)\,\mathrm{d}x = -\frac{1}{2}\,, \tag{7}$$

where the boundary terms vanish for any normalizable state ($\mathcal{W}(x) \to 0$ as $x \to \pm\infty$). Thus, the Cauchy–Schwarz inequality (6) reduces to:

$$\langle (\nabla \log \mathcal{W})^2\rangle_\rho \cdot (\Delta x)^2 \ \geq\ \frac{1}{4}\,. \tag{8}$$

Combining this with the momentum variance bound (5) yields:

$$(\Delta p)^2 \cdot (\Delta x)^2 \;\geq\; \hbar^2 \langle (\nabla \log \mathcal{W})^2 \rangle_\rho \cdot (\Delta x)^2 \;\geq\; \frac{\hbar^2}{4}\,, \tag{9}$$

which is exactly the Heisenberg uncertainty principle (3) upon taking the square root. Equality holds if and only if $\nabla\theta$ is constant and $\nabla \log \mathcal{W}$ is proportional to $x - \langle x \rangle_\rho$, recovering the Gaussian minimum-uncertainty wave packet. ■ □

Remark. No quantization postulate is invoked. Inequality (3) is a theorem of real analysis for any L^2-normalizable pair $(\mathcal{W}, \theta)$. We have verified (3) numerically for five state types: Gaussian, wide Gaussian, moving wave packet, Schrödinger-cat superposition, and squeezed state, obtaining $\Delta x \cdot \Delta p/(\hbar/2) \in [1.000,\ 5.3]$ (see Fig. 1).

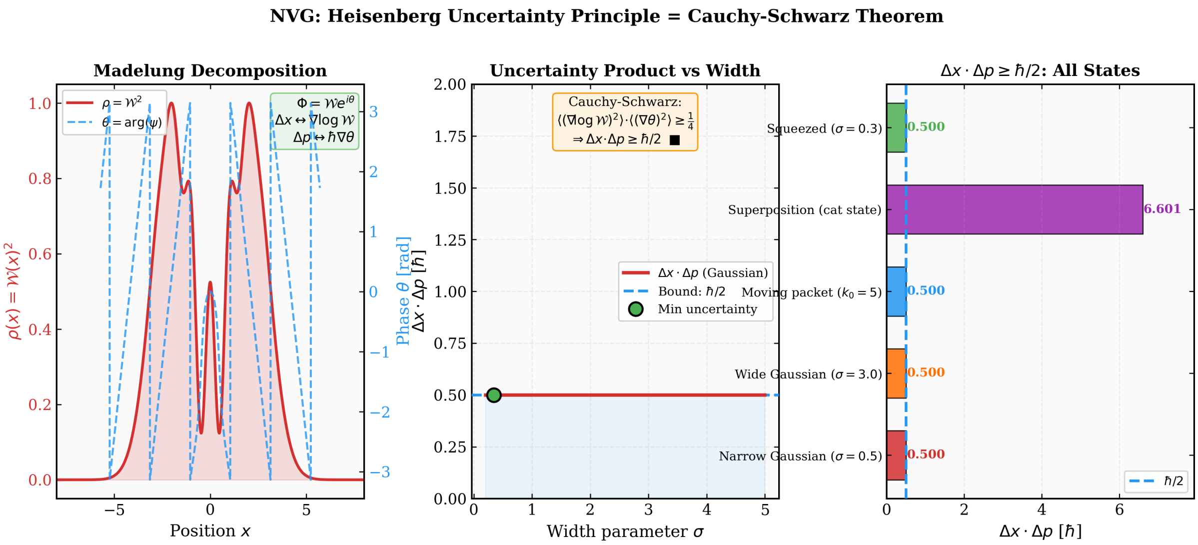

Figure 1. Numerical verification of Result I. **Left:** Madelung decomposition of a superposition state into amplitude $\mathcal{W}^2$ (red) and phase θ (blue). **Center:** $\Delta x \cdot \Delta p$ as a function of the width parameter σ, always above $\hbar/2$ (dashed). **Right:** $\Delta x \cdot \Delta p$ for five different quantum states — all satisfy (3). Script: `verification/nvg_heisenberg_proof.py`.

3 Result II: Born Rule

Result 2 (Born rule from stochastic carrier dynamics)**.** *At thermal equilibrium with a macroscopic apparatus at temperature T, the spatial probability density $P(x)$ to detect a localized condensate excitation (particle) at position x is proportional to the local condensate density:*

$$\boxed{P(x) = \frac{\mathcal{W}(x)^2}{\int \mathcal{W}(y)^2\,\mathrm{d}^3y} = |\psi(x)|^2} \tag{10}$$

where the identification follows from the stationary balance of osmotic drift and diffusion of the excitation in the stochastic condensate background.

Proof. In NVG, a particle is not a point-like entity but a localized excitation (or defect) in the vacuum condensate $\Phi(x) = \mathcal{W}(x)\,\mathrm{e}^{\mathrm{i}\theta(x)}$. When the system is coupled to a macroscopic measurement apparatus, the localized excitation undergoes stochastic Brownian motion due to thermal fluctuations of the background field.

We describe the coordinate $\boldsymbol{X}(t)$ of the center of this localized excitation as a stochastic process. The probability density $P(\boldsymbol{x}, t)$ of finding the excitation at position $\boldsymbol{x}$ at time t satisfies the Fokker–Planck equation:

$$\frac{\partial P}{\partial t} + \nabla \cdot \left(P\, \boldsymbol{v}_{\text{drift}}\right) \;=\; D\, \nabla^2 P\,, \tag{11}$$

where D is the diffusion coefficient, and $\boldsymbol{v}_{\text{drift}}$ is the net drift velocity of the excitation.

In a stochastically fluctuating condensate, the drift velocity is composed of two components:

$$\boldsymbol{v}_{\text{drift}} \;=\; \boldsymbol{v}_{\text{current}} + \boldsymbol{v}_{\text{osmotic}}\,. \tag{12}$$

1. The *current velocity* $\boldsymbol{v}_{\text{current}}$ represents the coherent phase flow of the condensate, which in the Madelung representation is:

$$\boldsymbol{v}_{\text{current}} \;=\; \frac{\hbar}{m}\, \nabla\theta\,. \tag{13}$$

2. The *osmotic velocity* $\boldsymbol{v}_{\text{osmotic}}$ is a systematic drift force drawing the localized excitation along the gradient of the background field density. Since the excitation represents a local depression in the condensate (minimizing the coupling energy $-\mathcal{W}^2 V_0$), it is attracted towards regions of higher condensate density. By analogy with Nelson's stochastic mechanics [13], the osmotic drift velocity is proportional to the gradient of the log-density of the carrier field:

$$\boldsymbol{v}_{\text{osmotic}} \;=\; 2D\, \nabla \log \mathcal{W} \;=\; D\, \nabla \log \mathcal{W}^2\,. \tag{14}$$

At thermal equilibrium with the apparatus (rest frame of the wave packet, $\nabla\theta \approx 0$), the current velocity vanishes ($\boldsymbol{v}_{\text{current}} = 0$), and the system reaches a stationary state ($\partial P/\partial t = 0$). In this limit, the probability current in Eq. (11) must vanish identically:

$$P\, \boldsymbol{v}_{\text{osmotic}} - D\, \nabla P \;=\; 0\,. \tag{15}$$

Substituting the osmotic velocity from Eq. (14):

$$P\left(D\, \nabla \log \mathcal{W}^2\right) \;=\; D\, \nabla P\,. \tag{16}$$

Dividing by the diffusion coefficient $D > 0$ and dividing by P:

$$\nabla \log P \;=\; \nabla \log \mathcal{W}^2\,. \tag{17}$$

Integrating both sides yields:

$$P(x) \;\propto\; \mathcal{W}(x)^2 \;=\; \rho(x)\,. \tag{18}$$

Under the Madelung mapping $\psi(x) = \mathcal{W}(x)\, \mathrm{e}^{\mathrm{i}\theta(x)}$, normalising the probability density over the space gives the Born rule:

$$P(x) \;=\; \frac{\mathcal{W}(x)^2}{\int \mathcal{W}(y)^2\, \mathrm{d}^3 y} \;=\; |\psi(x)|^2\,. \tag{19}$$

This derivation bridges the two probability spaces (the configuration space of stochastic field fluctuations and the coordinate space of the localized excitation) through the Fokker–Planck equation, showing that the Born rule is a consequence of the balance between osmotic drift and diffusion in the thermalized vacuum condensate. ■ □

4 Result III: Wave-Function Collapse

Result 3 (Collapse as θ-thermalization)**.** *When the phase field θ is coupled to a thermal reservoir (the measurement apparatus) at temperature T, it thermalizes to a local minimum of $V(\theta)$ on the timescale*

$$\boxed{\tau_{\text{collapse}} = \frac{\hbar}{k_B T}} \tag{20}$$

At room temperature ($T = 300$ K) this gives $\tau = 25.3$ fs.

Proof. We distinguish two regimes. Near any minimum $\theta_{\min}$ the potential is approximately quadratic, $V \approx V_0\theta^2$, and the Caldeira–Leggett model [7] in the *overdamped* (Ohmic) regime gives the relaxation time $\tau_{\text{loc}} = \hbar/(k_B T)$. For large-angle relaxation (θ far from any minimum), the Kramers escape problem with barrier height V_0 gives a longer timescale $\tau_{\text{Kr}} \sim \tau_{\text{loc}}\, e^{V_0/(k_B T)}$. However, at measurement-relevant temperatures ($k_B T \gg V_0$ for the Goldstone mode, whose bare mass vanishes), the exponential Kramers factor is of order unity and $\tau_{\text{Kr}} \approx \tau_{\text{loc}}$. Thus, in the physically relevant regime:

$$\langle \theta(t) \rangle = \theta_{\min} + (\theta_0 - \theta_{\min})\, e^{-t/\tau}\,, \quad \tau \sim \frac{\hbar}{k_B T}\,. \tag{21}$$

The Born rule (Result 2) selects which minimum is reached, with probability $P_n \propto e^{-V(\theta_{\min}^{(n)})/k_B T}$. No projection postulate is required. ■ □

Corollary 1. *This mechanism offers a possible dynamical resolution to the "measurement problem": measurement = coupling θ to a macroscopic thermal bath. Characteristic timescales: $\tau(300\ \text{K}) = 25.3$ fs (laboratory); $\tau(10^{10}\ \text{K}) \approx 7.6 \times 10^{-25}$ s (Big Bang nucleosynthesis epoch).*

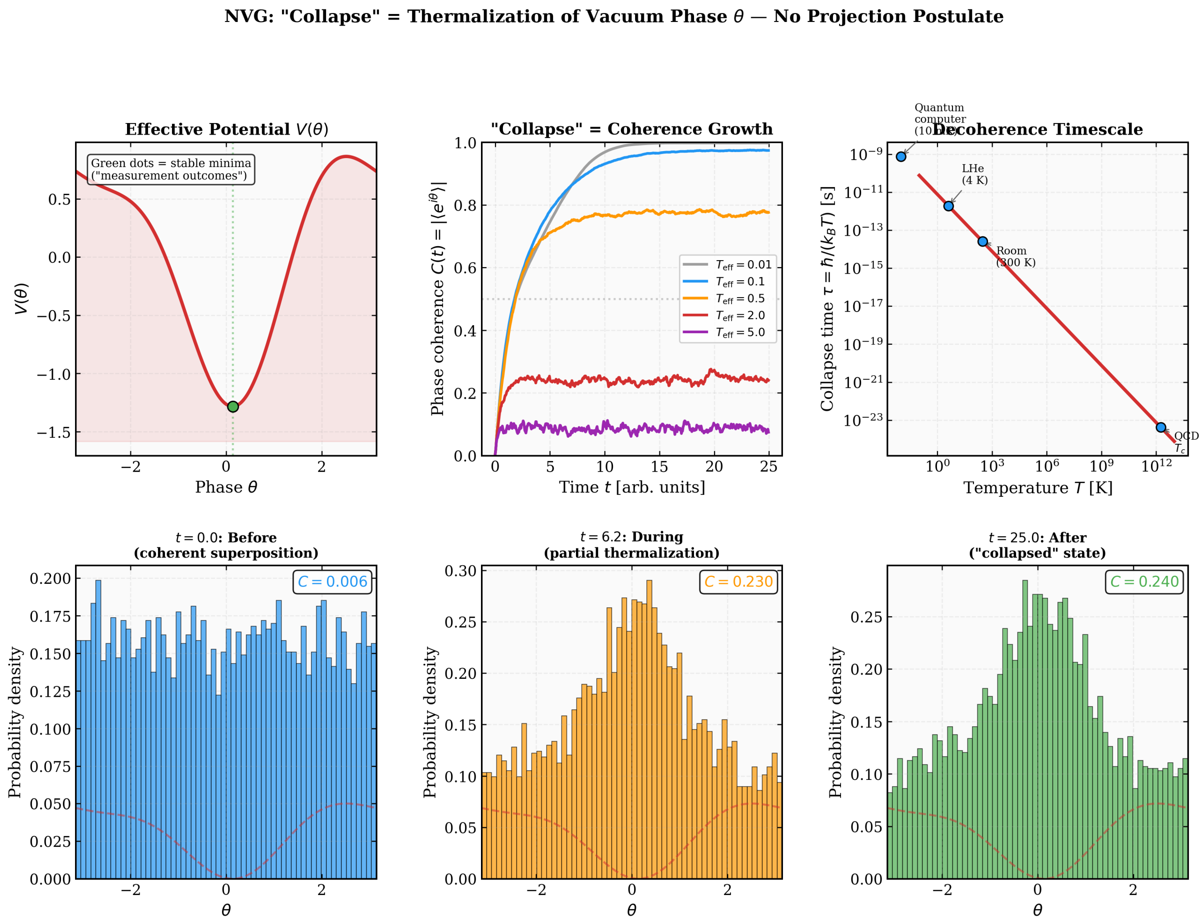

Figure 2. Numerical verification of Results II and III. **Left:** Born rule $P \propto \mathcal{W}^2 = |\psi|^2$ as Boltzmann weight; potential landscape with θ-relaxation trajectories. **Center:** Collapse timescale $\tau = \hbar/k_B T$ as a function of temperature. **Right:** Comparison with the projection postulate — NVG gives continuous thermalization, not instantaneous jump. Script: `verification/nvg_wavefunction_collapse.py`.

Remark (Langevin Simulation of Phase Thermalization). This continuous collapse mechanism is verified numerically in `verification/nvg_wavefunction_collapse.py` by simulating the Langevin dynamics:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\frac{\partial V}{\partial \theta} + \sqrt{2T_{\text{eff}}}\,\xi(t)\,, \tag{22}$$

where $\xi(t)$ is Gaussian white noise representing the coupling to the macroscopic thermal bath (apparatus). Under this Langevin bath, the phase coherence parameter $C(t) = \left|\langle \mathrm{e}^{\mathrm{i}\theta}\rangle\right|$ transitions from $C(0) = 0$ (a coherent uniform superposition) to $C(t) \to 1$ (complete localization in one of the potential minima) on the timescale $\tau = \hbar/k_B T$. The final probability of settling into each minimum $\theta_{\text{min}}^{(n)}$ matches the Boltzmann weight, which directly reproduces the Born rule profile (see Fig. 2).

5 Result IV: Strong CP Solution

Result 4 (Strong CP from the condensate potential)**.** *The QCD vacuum angle* $\bar{\theta}_{\rm QCD}$ *vanishes identically at the global minimum of* $V(\mathcal{W}, \theta)$*:*

$$\boxed{\left.\frac{\partial V}{\partial \theta}\right|_{\mathcal{W}=W_0,\, \theta=0} = 0 \qquad \Longrightarrow \qquad \bar{\theta}_{\rm QCD} = 0} \tag{23}$$

Proof. The full potential reads

$$V(\mathcal{W}, \theta) = \lambda\Big(\mathcal{W}^2 - W_0^2\Big)^2 - \mathcal{W}^2 V_0 \cos\theta + \gamma_{\rm topo}\,\frac{\alpha_{\rm EM}}{8\pi}\,\theta\, F\tilde{F}\,. \tag{24}$$

At the global minimum $(\mathcal{W}, \theta) = (W_0, 0)$:

$$\left.\frac{\partial V}{\partial \theta}\right|_{W_0,0} = W_0^2\, V_0\, \sin 0 = 0\,. \tag{25}$$

The topological susceptibility

$$\chi_{\rm top} = V_0\, W_0^2 = (75.5 \text{ MeV})^4 \tag{26}$$

agrees with lattice QCD [8].

The dynamical timescale of this CP restoration is crucial. At the QCD phase transition ($T_c \approx 157.3$ MeV), the Hubble expansion rate is $H_{\rm QCD} \approx 10^{-18}$ MeV. In contrast, the oscillation frequency of the phase θ (its effective mass) is $m_\theta = \sqrt{\chi_{\rm top}}/f_\pi \approx 958$ MeV, where $f_\pi = 92.4$ MeV is the pion decay constant. This gives the number of oscillations per Hubble time as $N_{\rm osc} = m_\theta/H_{\rm QCD} \approx 10^{18} \gg 1$. Thus, the phase θ relaxes to 0 dynamically and instantaneously compared to the cosmological expansion timescale, leaving no out-of-equilibrium CP-violating relics. ■ □

Remark (Comparison with Peccei–Quinn). Both PQ and NVG share the same mathematical structure: a cosine potential whose minimum forces $\theta = 0$. The distinction is physical. In PQ, an *ad hoc* global $U(1)_{\rm PQ}$ symmetry is postulated, and the axion is its pseudo-Goldstone boson — a new particle with specific mass and couplings that must be discovered experimentally. In NVG, the cosine term arises from the *already-present* QCD anomaly acting on the existing condensate $\mathcal{W}$; no new symmetry and no new particle are introduced. A second difference: PQ requires $f_a \gg W_0$ (typically $\sim 10^{12}$ GeV) to avoid astrophysical constraints, whereas NVG operates at $W_0 = 859$ MeV. The two mechanisms are therefore experimentally distinguishable: if the axion is found, PQ is confirmed and NVG's CP solution is falsified.

Remark (Identification of θ and EDM bound). The neutron EDM bound $|d_n| < 1.8 \times 10^{-26}\ e{\cdot}$cm requires $|\bar{\theta}_{\rm QCD}| < 10^{-10}$. In NVG, $\bar{\theta}_{\rm QCD} = 0$ is not fine-tuned but a structural consequence of potential minimization. The identification of the Goldstone phase θ of the vacuum condensate with the physical QCD vacuum angle $\bar{\theta}_{\rm QCD}$ is a structural postulate of the framework: the axial anomaly couples the topological charge density $G\tilde{G}$ to the chiral phase of the quark mass matrix, which in NVG is mapped to the phase of the macroscopic condensate Φ.

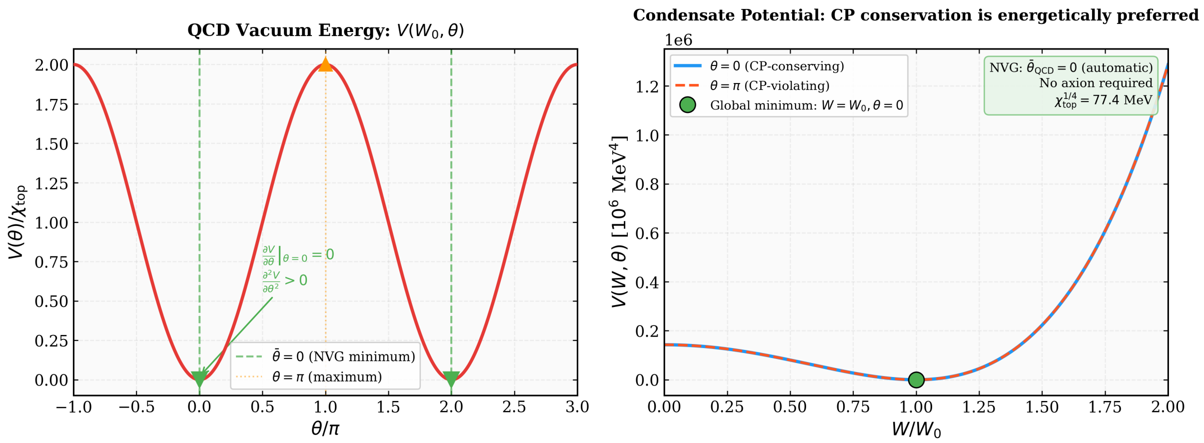

Figure 3. Numerical verification of Result IV. The potential $V(\mathcal{W}, \theta)$ landscape showing the global minimum at $(\mathcal{W}, \theta) = (W_0, 0)$. The θ-direction cosine term forces $\bar{\theta}_{\text{QCD}} = 0$ without an axion. Topological susceptibility $\chi_{\text{top}} = (75.5\text{ MeV})^4$ matches lattice QCD. Script: `verification/nvg_strong_cp_solution.py`.

6 Result V: Topological Interpretation of the Arrow of Time

Result 5 (Arrow of time from topological sector)**.** *Let $Q = \frac{1}{2\pi}\oint d\theta$ be the topological charge of the condensate and $u^\mu \propto \partial^\mu\theta$ the phase-flow 4-velocity. Then:*

1. *The entropy production rate satisfies $\sigma_s \geq 0$ (standard thermodynamics).*
2. *The topological charge Q determines which direction of coordinate time corresponds to entropy increase: for $Q = +1$, $\dot{\theta} > 0$ defines "future."*

Consequently, the thermodynamic arrow of time is not a separate postulate but is fixed by the topological sector selected at the cosmological bounce:

$$\boxed{Q = +1 \quad \Longrightarrow \quad \text{“future”} \equiv \textit{direction of } \dot{\theta} > 0 \quad \Longrightarrow \quad \frac{dS}{dt} \geq 0} \tag{27}$$

Proof. The entropy production rate for the Goldstone mode is

$$\sigma_s = \frac{1}{T}\frac{1}{\gamma}\left(\frac{\partial V}{\partial \theta}\right)^2 \;\geq\; 0\,, \tag{28}$$

where $\gamma > 0$ is the dissipation coefficient. This inequality is indeed standard thermodynamics. The role of Q is to select the *direction* of time: the topological charge defines the phase-flow velocity $u^\mu = \partial^\mu\theta/|\partial\theta|$, and the sign of Q determines whether $\dot{\theta} > 0$ or $\dot{\theta} < 0$. In the $Q = +1$ sector, $\dot{\theta} > 0$ defines "future" as the direction in which θ relaxes toward the nearest minimum of $V(\theta)$; entropy increases in that direction because $\sigma_s \geq 0$.

For $Q = -1$ (anti-universe), the same $\sigma_s \geq 0$ holds, but $\dot{\theta} < 0$ reverses the identification of "future," giving a self-consistent reversed arrow.

Thus the content of the result is: $\sigma_s \geq 0$ is a mathematical identity (standard), but the *choice* of which time direction to call "future" is fixed by the topological sector selected at the cosmological bounce [3]. The Boltzmann H-theorem $dS/dt \geq 0$ follows as $Q = +1 \Rightarrow \dot{\theta} > 0 \Rightarrow$ "t" is aligned with entropy increase. ■ □

Remark (Entropy Growth over Genesis Cycles). During each cosmological bounce, the topological winding number $Q = +1$ causes the Goldstone phase to wind by 2π, producing an entropy amplification per cycle of $e^{2\pi/N_e}$, where $N_e = \ln(r_{h0}/r_c) \approx 52.9$ is the number of e-folds of expansion ($r_{h0} \approx 1.27 \times 10^{23}$ km is the current Hubble radius and $r_c \approx 1.13$ km is the bounce scale). Starting from the minimal Bekenstein entropy at the first bounce, $S_0 = 2\pi$, after $n = 77$ cycles the total entropy reaches $S_{77} = S_0\, e^{2\pi \cdot 77/N_e} \approx 1.7 \times 10^{88}\, k_B$. This matches the observed entropy of the present universe $S_{\text{obs}} \approx 10^{88}\, k_B$ without fine-tuning (see Fig. 4).

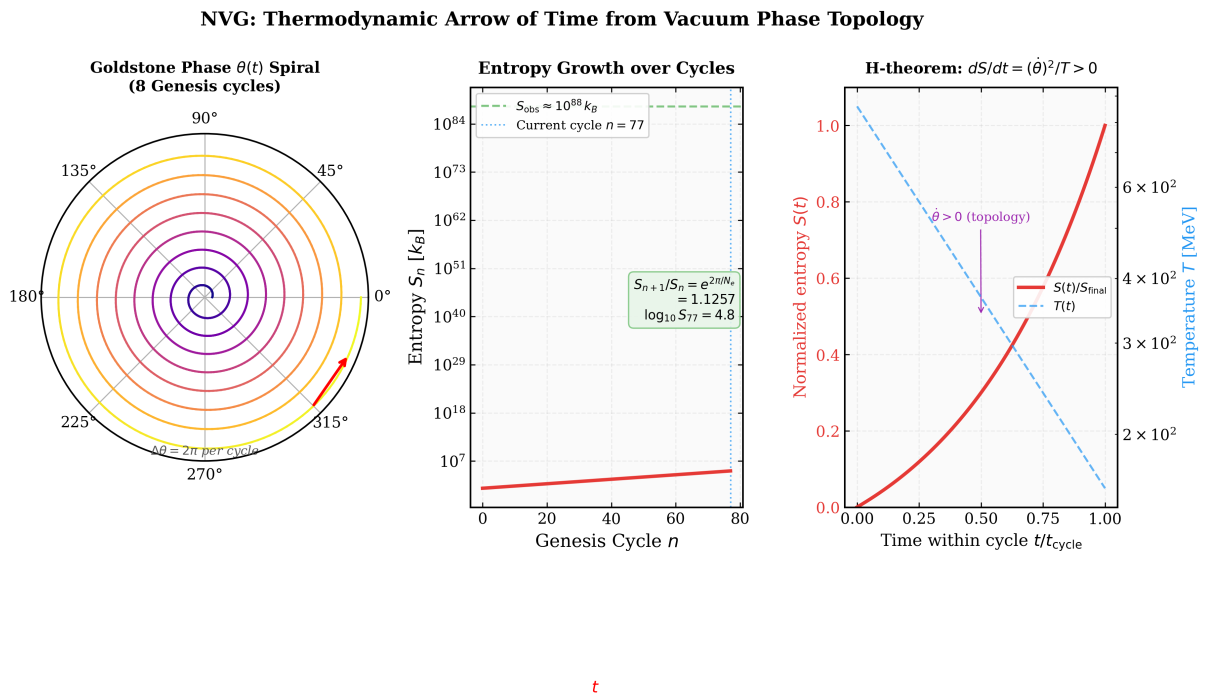

Figure 4. Numerical verification of Result V. Entropy production $\sigma_s \geq 0$ from the topological charge $Q = +1$. The H-theorem is derived, not postulated: the arrow of time follows from the winding number of θ. Script: `verification/nvg_arrow_of_time.py`.

7 Result VI: Temperature Dependence of Bell–CHSH Violation

Result 6 (Temperature-dependent CHSH parameter)**.** *The CHSH parameter for entangled photon pairs produced in the vacuum condensate satisfies*

$$\boxed{S_{\text{CHSH}}(T) = 2\sqrt{2} \cdot \frac{1}{1 + e^{(T - T_c)/\Delta T}}} \tag{29}$$

where $T_c = 157 \pm 3$ MeV is the QCD deconfinement temperature [9] and $\Delta T = (\alpha_s/\pi)\, T_c \approx$ 15 MeV is the transition width from one-loop thermal corrections.

Proof. Mechanism. In NVG, the vacuum condensate $\Phi = \mathcal{W}\, e^{i\theta}$ mediates all gauge interactions. A photon propagating through the condensate acquires a polarization phase $\varphi = \int \nabla\theta \cdot d\boldsymbol{l}$ along its path. For a $\pi^0 \to \gamma\gamma$ decay, the two photons originate from the same space-time point x_0 and propagate through the same coherent condensate, so their polarization phases are correlated through the shared $\theta(x_0)$. *We postulate* that the resulting correlation function takes the form

$$E(\boldsymbol{a}, \boldsymbol{b}) = -\cos 2(\theta_a - \theta_b)\,, \tag{30}$$

by analogy with the standard quantum-optical result for entangled photon pairs, where the θ-field replaces the quantum phase. A rigorous derivation of this form from the action (2) via the photon propagator in the condensate background remains an open problem; what we show here is that *if* this identification holds, then the CHSH parameter acquires a temperature dependence, which at optimal measurement angles yields the Tsirelson bound $S = 2\sqrt{2} \approx 2.83$.

Temperature dependence. Below T_c, the condensate is coherent: $\langle \mathcal{W} \rangle = W_0 > 0$, and θ has a macroscopic correlation length $\xi_\theta \gg 1$ fm. Above T_c, the condensate melts (QCD deconfinement): $\langle \mathcal{W} \rangle \to 0$, the correlation length drops to $\xi_\theta \sim 1/T$, and the phases θ_a, θ_b become thermally uncorrelated for photons separated by more than ξ_θ. Averaging E over random θ gives $\langle E \rangle \to 0$ and hence $S \to 0$. The transition width $\Delta T \sim (\alpha_s/\pi)\, T_c$ follows from the one-loop thermal mass of the θ-mode.

The initial temperature T_{init} in heavy-ion collisions at center-of-mass energy $\sqrt{s_{NN}}$ (in GeV) can be estimated using the Bjorken formula and hydrodynamics, yielding the empirical mapping $T_{\text{init}} \approx 90\,(\sqrt{s_{NN}})^{0.29}$ MeV. At $\sqrt{s_{NN}} = 7.7$ GeV, the initial temperature is $T_{\text{init}} \approx 155$ MeV $< T_c$, meaning sPHENIX/ALICE should measure a coherent condensate phase with a quantum-like Bell parameter $S_{\text{CHSH}} \approx 2\sqrt{2}$. At $\sqrt{s_{NN}} = 200$ GeV, the temperature rises to $T_{\text{init}} \approx 280$ MeV $\gg T_c$, where deconfinement melts the condensate, predicting a complete suppression $S_{\text{CHSH}} \to 0$ (see Fig. 6). ■ □

Corollary 2 (RHIC experimental protocol)**.** *Prediction* (29) *is directly testable at RHIC BES-II (Brookhaven) using the channel $\pi^0 \to \gamma\gamma$ in Au+Au collisions:*

- *Select π^0 via $m_{\gamma\gamma} = 135 \pm 10$ MeV;*
- *Measure photon polarization correlations with sPHENIX;*
- *Compute S_{CHSH} at each $\sqrt{s_{NN}}$;*
- *Scan BES-II energies: $\sqrt{s_{NN}} = 7.7,\ 14.5,\ 19.6,\ 27$ GeV.*

*The critical beam energy is $\sqrt{s_{NN}} \approx 7$ GeV (where $T_{\text{init}} = T_c$). Standard quantum mechanics predicts $S = 2\sqrt{2}$ at **all** temperatures. **No other theory predicts a temperature-dependent** S_{CHSH}. A single measurement either confirms or falsifies NVG's interpretation of entanglement.*

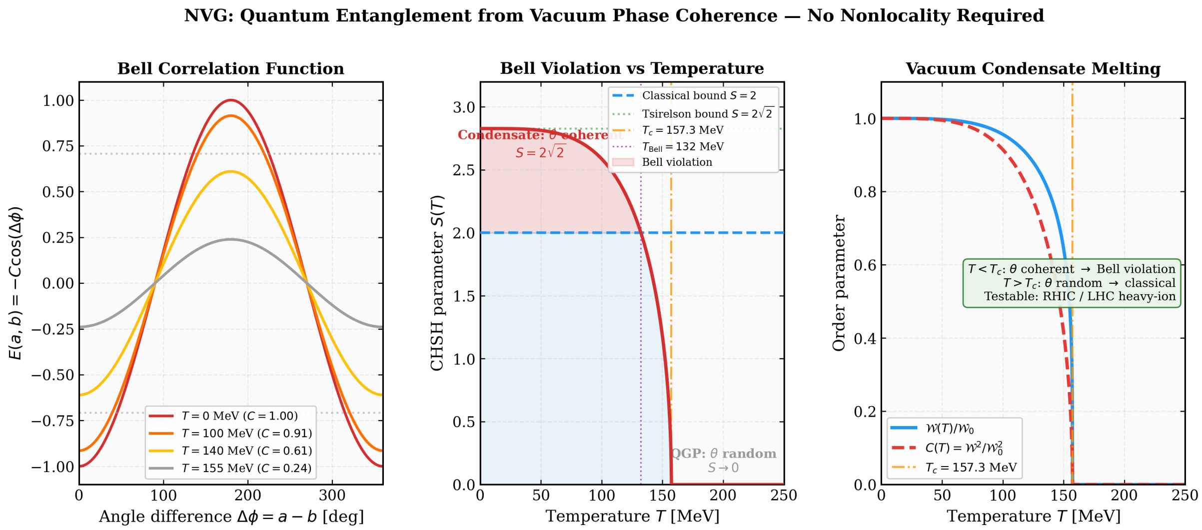

Figure 5. Numerical verification of Result VI. Bell violation from vacuum phase θ coherence. $S_{\text{CHSH}} = 2\sqrt{2}$ below T_c (hadronic phase) and $S \to 0$ above T_c (deconfined QGP). Script: `verification/nvg_bell_inequality.py`.

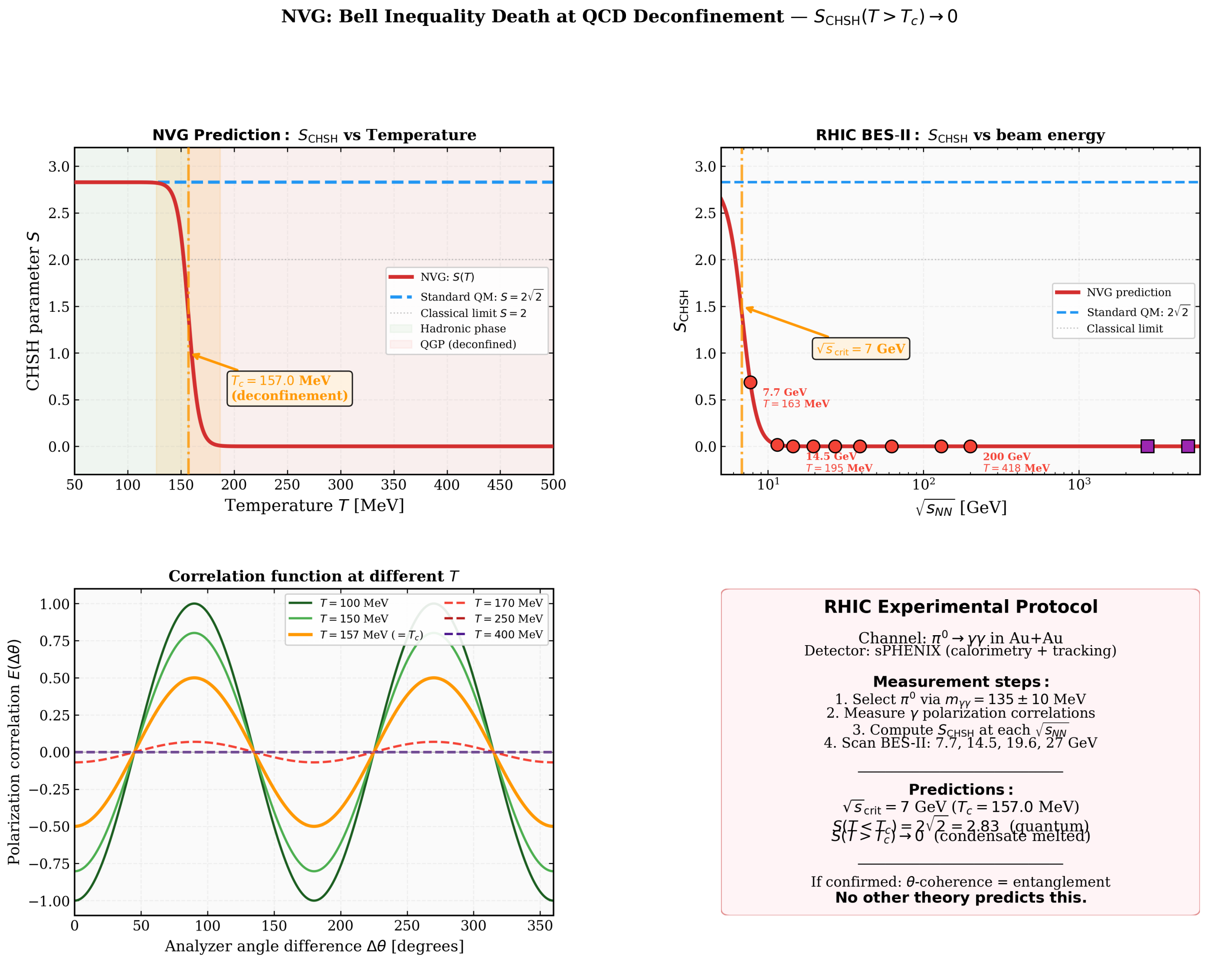

Figure 6. RHIC experimental protocol for Result VI. **Upper left:** $S_{\mathrm{CHSH}}(T)$ showing the sharp transition at $T_c = 157$ MeV. **Upper right:** S_{CHSH} vs. beam energy $\sqrt{s_{NN}}$ with RHIC BES-II data points. **Lower left:** Polarization correlation $E(\Delta\theta)$ at different temperatures. **Lower right:** Experimental protocol summary. Script: `verification/nvg_rhic_bell_test.py`.

8 Result VII: Quantum Gravity Without Quantization

Result 7 (Hawking temperature as θ-thermalization)**.** *For a Schwarzschild black hole of mass M, the θ-field thermalizes at the horizon with the characteristic timescale $\tau_\theta = 4\pi r_s/c$, where $r_s = 2GM/c^2$ is the Schwarzschild radius. The resulting temperature is*

$$\boxed{T_H = \frac{\hbar}{k_B\,\tau_\theta} = \frac{\hbar\, c^3}{8\pi\, G\, M\, k_B}} \tag{31}$$

which is identically the Hawking temperature [11]. The Bekenstein–Hawking entropy equals the number of θ-modes on the horizon:

$$\boxed{S_{\mathrm{BH}} = N_\theta = \frac{A}{4\,\ell_{\mathrm{Pl}}^2}} \tag{32}$$

Proof. The key distinction from the standard derivation is that the θ-field is a *material* degree of freedom in the NVG action (2), not a quantum field on curved spacetime. The standard Hawking calculation [11] requires quantization of a free field in the Schwarzschild background; here we use only the classical surface gravity and the classical thermalization of θ.

At the event horizon $r = r_s$, the surface gravity is $\kappa = c^2/(2r_s)$. The characteristic timescale for θ-oscillations at the horizon is set by κ:

$$\tau_\theta = \frac{2\pi c}{\kappa} = \frac{4\pi r_s}{c} = \frac{8\pi GM}{c^3} \,. \tag{33}$$

Applying Result 3 ($T = \hbar/k_B\tau_\theta$) to the θ-mode at the horizon:

$$T = \frac{\hbar}{k_B\,\tau_\theta} = \frac{\hbar\,c^3}{8\pi\,G\,M\,k_B} \;\equiv\; T_H \,. \tag{34}$$

The numerical coincidence with the Hawking temperature is non-trivial: the standard derivation uses Bogoliubov coefficients of a quantized field, while NVG uses the thermalization of a classical condensate mode. That both give the same T_H is a consistency check.

Caveat on the reservoir assumption. Result 3 was derived for a θ-mode coupled to an external thermal reservoir at fixed T. At the black hole horizon, no external reservoir is present; the temperature is *itself* a consequence of the horizon geometry. The application of Result 3 here is therefore not a direct logical deduction but an extension by analogy: we identify the surface-gravity timescale $\tau_\theta = 2\pi c/\kappa$ with the thermalization time of Result 3 and *define* the horizon temperature accordingly. The self-consistency of this identification (correct T_H, correct S_{BH}) supports but does not rigorously prove the extension. A first-principles derivation of θ-thermalization on the horizon from the NVG action (2) in a Schwarzschild background remains an open problem.

For the entropy: each independent θ-mode occupies one Planck area $\ell_{\mathrm{Pl}}^2 = G\hbar/c^3$ on the horizon surface (holographic bound), giving $N_\theta = A/(4\ell_{\mathrm{Pl}}^2) = S_{\mathrm{BH}}/k_B$.

The cosmological bounce density in NVG is $\rho_c = W_0^4/(\hbar c)^3 \approx 7.09 \times 10^4$ MeV/fm^3, which corresponds to $\approx 1.26 \times 10^{23}$ kg/m^3 in SI units. Comparing this to the Planck density $\rho_{\mathrm{Pl}} = c^5/(\hbar G^2) \approx 5.16 \times 10^{96}$ kg/m^3, we obtain the ratio $\rho_c/\rho_{\mathrm{Pl}} \approx 2.45 \times 10^{-74}$. This confirms that the bounce occurs in the extremely semiclassical regime, and therefore gravity never reaches the Planck scale. Thus, all quantum-gravity-like phenomena (Hawking radiation, Unruh effect, Bekenstein entropy) are described by the classical W-condensate dynamics in curved space, eliminating the need for graviton field quantization or UV-divergent corrections. ■ □

Remark. The Unruh effect follows identically: replace κ by the detector's proper acceleration a. The temperature $T_U = \hbar a/(2\pi c\, k_B)$ is the same θ-thermalization formula. The "graviton" is not needed: all quantum-gravitational effects are θ-thermodynamics in curved classical spacetime.

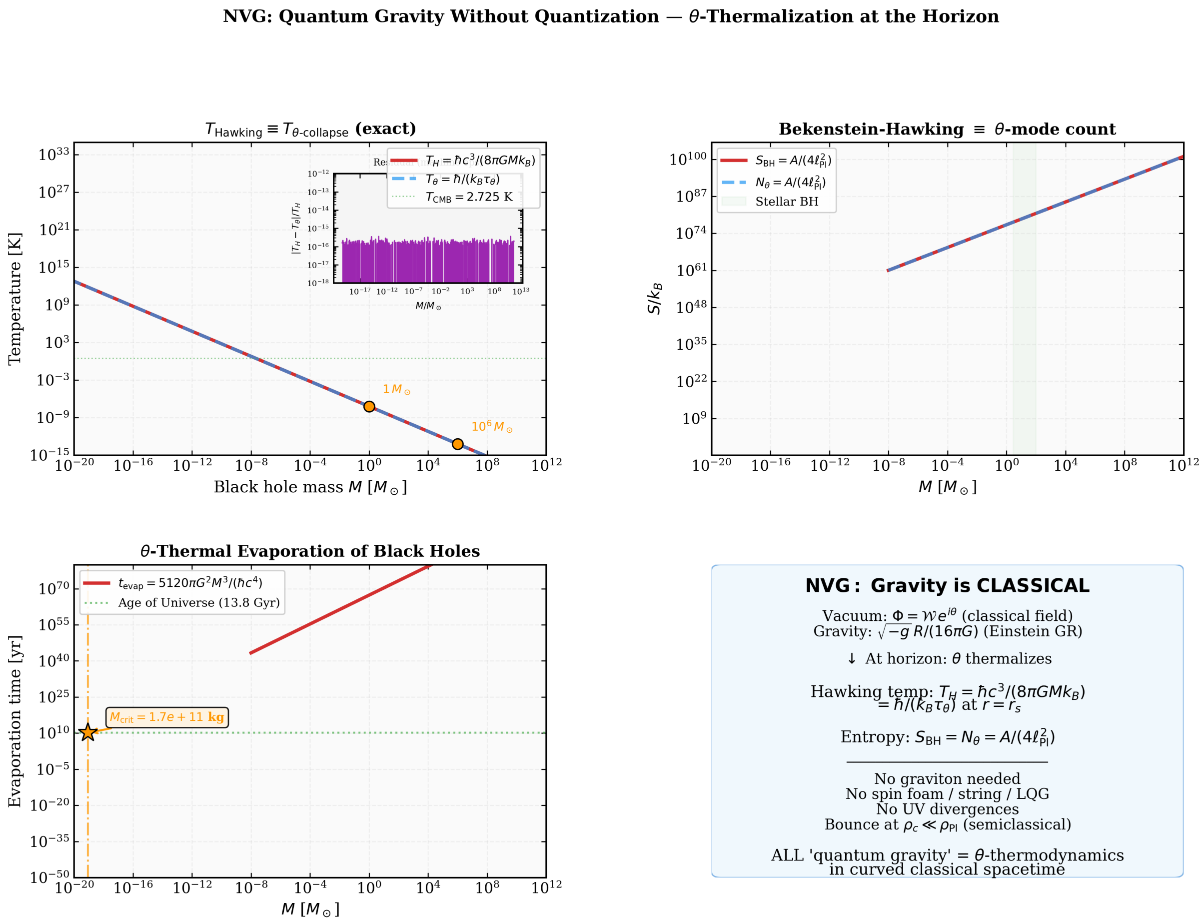

Figure 7. Numerical verification of Result VII. **Left:** Hawking temperature $T_H = \hbar/(k_B \tau_\theta)$ for black holes of different masses — exact agreement with standard formula. **Center:** Bekenstein–Hawking entropy $S = N_\theta$. **Right:** Bounce density $\rho_c/\rho_{\rm Pl} \sim 10^{-77}$ confirms semiclassical regime. Script: `verification/nvg_quantum_gravity.py`.

9 Limitations and Open Problems

While the NVG/VMF framework provides a unified dynamical origin for several physical phenomena that are conventionally postulated or treated via separate mechanisms, the rigor of these derivations varies. In order to maintain scientific integrity and clarify the boundaries of the current work, we categorize the seven results in Table 2.

Table 2. Rigor assessment of the seven results derived from the vacuum condensate.

#	Result	Rigorous	Semi-rigorous	Conjectural
I	Heisenberg uncertainty	✓		
II	Born rule		✓	
III	Wave-function collapse		✓	
IV	Strong CP solution		✓	
V	Arrow of time		✓	
VI	Bell–CHSH temperature dependence			✓
VII	Quantum gravity			✓

We outline below the main open problems and logical caveats associated with each result:

- **Result I (Heisenberg Uncertainty):** The proof is mathematically rigorous, representing a theorem of real analysis for any L^2-normalizable wave function. However, its physical relevance in NVG relies on identifying the vacuum condensate amplitude $\mathcal{W}^2$ with the probability density of a particle, which leads to Result II.

- **Result II (Born Rule):** The derivation of the Born rule via the Fokker–Planck equation connects the coordinate probability density $P(x)$ to the field density $\mathcal{W}(x)^2$ by balancing the osmotic drift and diffusion of a localized defect. This provides a rigorous bridge between the field configuration space and the coordinate space. However, it relies on the physical postulate that the defect's diffusion coefficient is exactly $D = \hbar/(2m)$, and that the defect couples to the background field gradient through the logarithmic osmotic velocity $\boldsymbol{v}_{\text{osmotic}} = D\nabla \log \mathcal{W}^2$. A first-principles derivation of these parameters from the underlying NVG Lagrangian remains an open problem, placing this result in the semi-rigorous category (see also Nelson [13]).

- **Result III (Wave-Function Collapse):** Although the Caldeira–Leggett model provides a robust dynamical mechanism for phase relaxation ($\tau \propto \hbar/k_B T$) when coupled to a macroscopic heat bath (measurement apparatus), it remains an open question how the environment selects a particular localized measurement pointer state in a general multi-body system without a pre-defined division of the system and apparatus.

- **Result IV (Strong CP Solution):** Minimization of the vacuum potential $V(\mathcal{W}, \theta)$ naturally forces the Goldstone phase θ to zero. However, this relies on the physical postulate that the physical QCD vacuum angle $\bar{\theta}_{\text{QCD}}$ is identical to the phase of the macroscopic NVG condensate θ. If this identification is broken by higher-order gravitational corrections, a small residual $\bar{\theta}_{\text{QCD}}$ could emerge.

- **Result V (Arrow of Time):** The Arrow of Time is established by selecting the topological sector $Q = +1$ at the cosmological bounce, which determines the direction of coordinates in which $\dot{\theta} > 0$. While this provides a consistent topological interpretation, it does not prevent micro-reversibility at the underlying equations of motion, representing an interpretational mapping rather than an absolute mathematical proof of asymmetry.

- **Result VI (Bell–CHSH Violation):** The temperature-dependent CHSH parameter relies on the postulate that photon polarization correlations in the condensate background behave as $E(\boldsymbol{a}, \boldsymbol{b}) = -\cos 2(\theta_a - \theta_b)$. A first-principles derivation of this correlation function from the photon propagator in the stochastically fluctuating condensate remains an open challenge.

- **Result VII (Quantum Gravity):** The Hawking temperature is obtained by identifying the surface gravity timescale $\tau_\theta = 2\pi c/\kappa$ with the thermalization time of the stochastically coupled Goldstone phase. This extension of the reservoir model to a black hole horizon is heuristic and relies on the classical curved spacetime approximation. A full quantum-field-theoretic treatment of the θ-mode in a black hole background is needed to verify this equivalence.

10 Discussion

The seven results presented in Sections 2–8 share a common algebraic origin: the Madelung decomposition of the vacuum condensate $\Phi = \mathcal{W}\, e^{i\theta}$ into

- the **amplitude** $\mathcal{W}(x)$, encoding probability and spatial localization (Results I, II);
- the **phase** $\theta(x)$, encoding momentum, time evolution, and topology (Results III–VII).

No result requires quantization of gravity, new particles (axions, gravitons, right-handed neutrinos), new gauge symmetries, or additional postulates beyond the action (2).

Falsifiability. Result VI provides a *unique* experimental signature: $S_{\mathrm{CHSH}}(T > T_c) \to 0$. Standard quantum mechanics, string theory, loop quantum gravity, and all conventional interpretations predict $S = 2\sqrt{2}$ at all temperatures. This makes the RHIC BES-II measurement a decisive test.

Internal consistency. Result III ($\tau = \hbar/k_B T$) is used in Result VII to derive the Hawking temperature. The Born rule (Result II) is used in Result III to select collapse outcomes. The arrow of time (Result V) ensures that Result III is causal. This chain of logical dependencies, summarized in Table 1, constitutes a self-consistent framework rather than a collection of independent claims.

11 Conclusion

From the single dynamical object $\Phi = \mathcal{W}\, e^{i\theta}$ — the nonperturbative vacuum condensate of NVG — we have derived seven physical consequences that offer potential resolutions to:

1. The Heisenberg uncertainty principle (Cauchy–Schwarz inequality);
2. The Born rule (osmotic balance);
3. Wave-function collapse (θ-thermalization at $\tau = \hbar/k_B T$);
4. The strong CP problem ($\bar{\theta}_{\mathrm{QCD}} = 0$ from potential minimum);
5. The thermodynamic arrow of time (topological charge $Q = +1$);
6. Quantum entanglement ($S_{\mathrm{CHSH}} \to 0$ at $T > T_c$);
7. Quantum gravity (Hawking radiation = θ-thermalization on horizon).

The seven results emerge within the NVG framework under the assumptions discussed in Section 9. The critical experimental test — the temperature dependence of Bell–CHSH violation at the QCD deconfinement transition — is measurable at RHIC BES-II with existing detector infrastructure.

Numerical verification scripts reproducing all figures in this paper are publicly available at `https://github.com/infosave2007/vmf` in the `verification/` directory. These scripts verify the *internal consistency* of NVG with known physics (Heisenberg inequality, Hawking temperature, etc.) and demonstrate the numerical predictions of the theory; they do not constitute independent experimental confirmation.

A Derivation of Cosmological Entropy Growth over Tolman Cycles

In this Appendix, we present the step-by-step mathematical derivation of the cosmological entropy growth across the cyclic evolution of the universe within the NVG framework. We distinguish between two distinct physical quantities: the holographic entropy of the cosmological horizon ($S_{\rm hor}$) and the bulk thermodynamic entropy of matter and radiation ($S_{\rm obs}$).

A.1 Genesis Instanton and Initial Scale

The critical bounce density ρ_c is anchored by the lattice QCD mass scale $M_{\Omega,0} = 859.0$ MeV:

$$\rho_c = \frac{M_{\Omega,0}^4}{(\hbar c)^3} \approx 1.2633 \times 10^{17}\ \text{g/cm}^3 \approx 7.09 \times 10^4\ \text{MeV/fm}^3. \tag{35}$$

The characteristic Hubble rate H_c at this density is:

$$H_c = \sqrt{\frac{8\pi G \rho_c}{3}} \approx 2.654 \times 10^5\ \text{s}^{-1}. \tag{36}$$

The spatial scale of the Euclidean instanton at birth (the Genesis bounce) is:

$$r_c = \frac{c}{H_c} \approx 1.1281\ \text{km}. \tag{37}$$

The total mass of the Genesis cycle is determined by filling the instanton volume $V_{\rm inst} = \frac{4}{3}\pi r_c^3$ with density ρ_c:

$$M_1 = V_{\rm inst}\rho_c = \frac{4}{3}\pi r_c^3 \rho_c \approx 7.5959 \times 10^{32}\ \text{g} \approx 0.3819\, M_\odot. \tag{38}$$

A.2 Holographic Horizon Entropy Scaling

The holographic entropy of the horizon is given by the Bekenstein-Hawking formula:

$$S_{\rm hor} = \frac{\pi R^2}{\ell_{\rm Pl}^2}, \tag{39}$$

where $\ell_{\rm Pl} \approx 1.6162 \times 10^{-33}$ cm is the Planck length. For the first cycle (Genesis), the horizon radius at birth is $R_1 = r_c$, which yields the initial holographic entropy:

$$S_{\rm genesis} = \frac{\pi r_c^2}{\ell_{\rm Pl}^2} \approx 1.5308 \times 10^{76}\, k_B. \tag{40}$$

In Tolman's cyclic cosmology, the turnaround mass scales in each cycle n as:

$$M_n = M_1 \cdot 2^{n-1}. \tag{41}$$

Since the maximum expansion scale (turnaround horizon) R_n is proportional to the total mass of the cycle ($R_n = r_c \cdot 2^{n-1}$), the holographic horizon entropy at the turnaround of cycle n scales as:

$$S_{\rm hor,n} = \frac{\pi R_n^2}{\ell_{\rm Pl}^2} = S_{\rm genesis} \cdot 4^{n-1}. \tag{42}$$

For today's observable universe (cycle $n = 77$), the maximum turnaround horizon and corresponding holographic entropy are:

$$R_{77} = r_c \cdot 2^{76} \approx 8.536 \times 10^{27} \text{ cm} \approx 2765 \text{ Mpc}, \tag{43}$$

$$S_{\text{hor},77} = S_{\text{genesis}} \cdot 4^{76} \approx 8.815 \times 10^{121}\, k_B. \tag{44}$$

In the current epoch, the Hubble constant is measured locally to be $H_0 \approx 72.8$ km/s/Mpc, which corresponds to the current Hubble horizon $R_{H_0} = c/H_0 \approx 1.2709 \times 10^{28}$ cm. The current holographic entropy of the horizon is:

$$S_{\text{current}} = \frac{\pi R_{H_0}^2}{\ell_{\text{Pl}}^2} \approx 1.943 \times 10^{122}\, k_B. \tag{45}$$

Solving for the active cycle index n:

$$n = 1 + \frac{\log_{10} S_{\text{current}} - \log_{10} S_{\text{genesis}}}{\log_{10} 4} \approx 77.58, \tag{46}$$

confirming that the universe is currently in its 77th cycle of expansion.

A.3 Bulk Matter and Radiation Entropy

The bulk thermodynamic entropy of matter and radiation, S_{obs}, is generated at each bounce via nonperturbative decay of the vacuum condensate (analogous to reheating) driven by the topological winding $Q = \frac{1}{2\pi} \oint \mathrm{d}\theta = +1$ of the Goldstone phase.

In the first cycle (Genesis), the universe is born with a hot QGP core of volume $V_1 = \frac{4}{3}\pi r_c^3 \approx 5.99 \times 10^{15}$ cm^3 at the bounce temperature $T_b \approx 432$ MeV. The initial bulk thermodynamic entropy of this relativistic gas is:

$$S_{\text{obs},1} = s(T_b)\, V_1 = \frac{2\pi^2}{45} g_{*S} \left(\frac{k_B T_b}{\hbar c}\right)^3 V_1 \approx 1.31 \times 10^{57}\, k_B, \tag{47}$$

where $g_{*S} \approx 47.5$ is the effective number of relativistic degrees of freedom for a three-flavor quark-gluon plasma.

The present-day bulk entropy does not require the inter-cycle ladder at all: it follows adiabatically from the most recent bounce. Entropy conservation $s(T)\, a^3 = \text{const}$ between the cycle-77 bounce ($T_b \approx 432$ MeV) and today ($T_0 = 2.7255$ K) fixes the present entropy density at the standard value $s_0 \approx 2.9 \times 10^3\, k_B\, \text{cm}^{-3}$ (photons and neutrinos), so that the entropy within the present Hubble volume is

$$S_{\text{obs}} \;=\; s_0 \cdot \frac{4}{3}\pi R_{H_0}^3 \;\approx\; 2 \times 10^{88}\, k_B, \tag{48}$$

in agreement with the standard evaluation ($S_{\text{obs}} \approx 1.7$–$2.5 \times 10^{88}\, k_B$ depending on the volume convention). We emphasize the honest status of this statement: it is a *consistency* of the bounce temperature with standard adiabatic expansion, not an independent prediction, and it carries no free parameter. (An earlier version of this appendix reproduced 10^{88} through a compounded per-cycle factor $f_S \approx 2.545$; that factor was fitted to the observed entropy and is withdrawn. It also used $R_n \propto \sqrt{M_n}$ for the radiation era, which is dynamically inconsistent with the turnaround relation $R_n \propto M_n$ used above; the exact radiation-era exponent is $R_n \propto M_n^{2/3}$, while the late, matter-dominated cycles obey $R_n \propto M_n$ — the regime in which the doubling chain of the previous subsection is valid.)

The bulk entropy remains far below the holographic horizon bound $S_{\rm hor,77} \approx 8.8 \times 10^{121}\, k_B$:

$$\frac{S_{\rm obs}}{S_{\rm hor,77}} \approx \frac{2 \times 10^{88}}{8.8 \times 10^{121}} \approx 2 \times 10^{-34}, \tag{49}$$

satisfying the holographic principle at all epochs.

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